ATLAS: Adaptive Trustworthy Language-Augmented Search

Multi-Turn Conversational Search Evaluation Report

Department of Creative Technologies, Air University, Islamabad
Course: Information Retrieval (IR) | Session: June 2026
Group Members: Asma Shoukat (241418), Amna-tuz-Zahra (241382)

1. Problem Statement & Dataset

Traditional Retrieval-Augmented Generation (RAG) models suffer from significant performance degradation in multi-turn conversational search. ATLAS addresses three core operational issues: **(1) Contextual Pronoun Dependency & Query Drift** — follow-up queries contain pronouns (*"it"*, *"they"*, *"them"*) that naive retrieval systems cannot resolve, while naively concatenating history causes vector drift; **(2) Hallucination Risk** — standard systems lack confidence filters and fabricate answers when evidence is insufficient; and **(3) Black-Box Opacity** — conventional neural pipelines provide no verifiable citations or audit trails.

ATLAS was evaluated on the **IBM Research Mt-RAG (SemEval 2026)** conversational benchmark spanning **366,479 plaintext passages** across four domains:

Domain	Document Source Type	Passages
Government (GOVT)	NASA Space Missions, FEMA Disaster Safety & Public Policy	49,607
Finance (FIQA)	Financial News Analyst Reports & Q&A Forums	61,022
Cloud (CLOUD)	IBM Cloudant Technical Manuals & Cloud Developer Docs	72,442
Wikipedia (CLAPNQ)	General Knowledge Factoid Q&A & World History	183,408
Total Corpus	**Multi-Domain Mt-RAG Database Pool**	**366,479**

Preprocessing: Source documents are sentence-chunked via regex segmenters; dense 384-dimensional vectors are generated using *all-MiniLM-L6-v2* SentenceTransformers; a FAISS Inner Product (IP) index is compiled. For CPU efficiency, an active local index of ~1,091 passages (250 per domain + gold reference documents from tasks 10–60) is used, achieving 0.3s retrieval latency.

2. Methodology — 7-Pillar Pipeline

Pillars 1 & 2: Conversational Understanding & Query Rewriting

ATLAS maintains a **Dual-Channel Context Tracker**: (1) a *User Subjects Channel* extracting noun phrases from user inputs (filtering noise words, truncating to last 3 terms), and (2) a *Proper Nouns Channel* tracking capitalized terms across all turns. When a query contains general pronouns (*"it"*, *"they"*, *"them"*, *"its"*, *"their"*) or is very short (<5 words), the system resolves the pronoun to the most recent subject (e.g., *"What causes them?"* → *"What causes wildfires?"*) or enriches short queries by appending context (e.g., *"Which is more important?"* → *"Which is more important regarding Market Cap and NAV?"*).

Pillar 3: Dual-Channel Hybrid Retrieval

ATLAS runs two parallel retrieval channels: *Lexical Search (BM25 Okapi)* evaluates term frequencies across the corpus and returns the top 50 keyword-matched hits; *Dense Vector Search (FAISS)* encodes the query into 384-D space and performs Inner Product search, returning the top 50 semantic matches.

Pillar 4: Reciprocal Rank Fusion (RRF)

Candidates from both channels are merged using: **RRF_Score(d) =** $\sum_m$ **1/(60 + Rank$_m$(d))**. Documents retrieved by both channels are marked *HYBRID*. The top 20 fused candidates are retained.

Pillar 5: Neural Cross-Encoder Reranking

The top 10 candidates are reranked using *ms-marco-MiniLM-L-6-v2*, a Cross-Encoder that processes the query and document text jointly via full cross-attention, producing a logit confidence score (range: –10.0 to +10.0).

Pillar 6: Anti-Hallucination Decision Agent

Four safety gates verify evidence sufficiency: (1) *Domain-Calibrated Thresholds* — logit must exceed per-domain thresholds (CLOUD: –2.0, GOVT: –1.5, CLAPNQ: –2.0, FIQA: +2.0); (2) *Score-Gap Ambiguity* — if the gap between top two candidates is <0.5 with logit <0, the query is flagged; (3) *Semantic Overlap* — at least 2 content words must overlap (1 for ≤2-word queries); (4) *Named-Entity Consistency* — 2+ missing capitalized entities trigger interception. Failure produces a safe fallback: *"I'm sorry, but I cannot find sufficient evidence..."*

Pillar 7: Extractive Sliding-Window Sentence Scorer

The approved passage is split into sentences, each scored by query-token overlap, and the contiguous 2–3 sentence window with the highest cumulative score (plus position bias toward earlier content) is extracted. This strictly extractive method guarantees zero generative hallucinations.

System Workflow

User Query → **Dual-Channel Tracker** → **Query Rewriter** → **BM25 + FAISS Retrieval** → **RRF Fusion** → **Cross-Encoder Reranking** → **Decision Agent Guards** → *(if passed)* → **Extractive Sentence Scorer** → **Output + Citations**

3. Evaluation Results

The ATLAS pipeline was stress-tested across domains. Results demonstrate accurate retrieval, pronoun resolution, short-query enrichment, and anti-hallucination interception:

Scenario A: Wildfire Follow-Up (GOVT Domain)

Turn 1: "Which state has more wildfires?"

Verdict: APPROVED (Confidence: 3.847, threshold: –1.5). **Output:** California wildfire data from 'CAL FIRE' citing 4.2 million acres burned in 2020. **Citation:** `2806a7d8f0775d28-14422-16441`

Turn 2: "What causes them?"

Rewritten: "What causes wildfires?" (pronoun "them" → "wildfires"). **Verdict:** APPROVED (3.363). **Output:** Lightning and human activities as primary causes, from 'Wildfire Hazard Mitigation'. **Citation:** `8bb77f30210c5f4b-277-2698`

Scenario B: Market Cap vs NAV (FIQA Domain)

Turn 1: "What's the difference between Market Cap and NAV?"

Verdict: APPROVED (7.431, threshold: 2.0). Retrieves definition of market capitalization vs. net asset value. **Citation:** `414940-0-474`

Turn 2: "Which is more important?"

Rewritten: "Which is more important regarding Market Cap and NAV?" **Verdict:** APPROVED (4.652). Retrieves fund valuation comparison context.

Scenario C: Out-of-Domain Interception

Query: "How do you bake a vanilla wedding cake with buttercream frosting?" (GOVT)

Verdict: INTERCEPTED. Reranker score: –10.161 (below –1.5 threshold); score gap: 0.272 (ambiguous); 1 matching term (below 2-term minimum). Safe fallback message returned: *"I'm sorry, but I cannot find sufficient evidence..."*

Scenario D: Named-Entity Guard (FIQA)

Query: "What is the NAV of Apple?"

Verdict: INTERCEPTED. Confidence: –0.856 (below +2.0 threshold). Named-entity "Apple" absent from retrieved passage (which discusses Amazon).

4. Limitations & Individual Contribution

Limitations: (1) *Extractive-Only Generation* — the sliding-window scorer cannot paraphrase or combine multiple documents; poorly structured source sentences may reduce readability. (2) *Short Context Window* — rapid topic shifts over 10+ turns may cause pronoun resolution to reference stale subjects. (3) *No Cross-Document Synthesis* — answers are drawn from a single passage; multi-document reasoning is unsupported. (4) *CPU Latency* — global-index retrieval (366,479 passages) requires GPU acceleration for real-time performance; CPU mode is optimized via active sub-indexing.

Individual Contributions

Asma Shoukat (241418): Core backend search architecture — hybrid retrieval engine (BM25 + FAISS), Reciprocal Rank Fusion merger, Cross-Encoder neural reranker integration, and domain safety threshold calibration.

Amna-tuz-Zahra (241382): Dialogue state tracking and UI development — dual-channel context tracker for coreference resolution, responsive glassmorphic web interface with live pipeline telemetry, and sliding-window sentence extractor for citation highlighting.